

AMIT KUMAR BEDAKA

Senior Industrial Robotics & Automation Engineer | CAD/CAM Software

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PhD-qualified Senior Industrial Robotics & Automation Engineer with 7+ years of experience developing CAD/CAM-based robotic software, intelligent automation and manufacturing systems. Expertise in C++, Python, OpenCASCADE, robot kinematics, offline programming, path planning, ROS2 and industrial robot simulation. Developed end-to-end solutions for robotic welding, automated inspection, CAD-driven robot path generation and manufacturing automation, connecting CAD geometry, planning algorithms, simulation and robot execution. Experienced in leading multidisciplinary R&D and translating engineering concepts into validated industrial prototypes.

EXPERIENCE

Head of Design

Sensable

📅 May 2024 – Present 📍 Barcelona

- Leading mechanical and product design activities for an assistive technology product, from requirements definition and concept development through CAD design, prototype iteration and engineering integration.
- Coordinating product architecture, component design and design modifications while balancing performance, usability, cost and implementation constraints.
- Working across design, electronics and software activities to resolve integration issues and support technical demonstrations and design reviews.

Lead Scientist / Marie Skłodowska-Curie Fellow

Chordata Motion

📅 May 2022 – May 2024 📍 Barcelona

- Led development and validation of a novel indoor positioning prototype for human motion capture, coordinating requirements, prototype integration, testing and technical evaluation.
- Developed IMU-based estimation using ZUPT, PDR and EKF and evaluated RF, VR-tracking and vSLAM approaches in collaboration with Chordata Motion, TecnoCampus Motion Capture Lab and the WiNE group at UOC.
- Managed project milestones and experimental validation against optical motion-capture reference systems, analyzing accuracy, repeatability and integration limitations.

Postdoctoral Researcher

National Taiwan University of Science and Technology

📅 April 2019 – May 2022 📍 Taipei, Taiwan

- Developed C++/OpenCASCADE CAD/CAM software for industrial robot offline programming and manufacturing automation.
- Built workflows connecting CAD geometry → feature/geometry processing → robot path generation → kinematics → simulation → robot execution.
- Developed robotic automation methods for welding, autonomous inspection and manufacturing, integrating industrial robots, CAD models and 3D sensing.
- Validated algorithms through robotic simulation and experimental systems using industrial robot environments.

Consultant (R&D Lead)

CENZ Automation

📅 June 2019 – Aug 2022 📍 Taipei, Taiwan

- Led R&D development of CAD/CAM and robotic

EDUCATION

Ph.D. in Mechanical Engineering (Robotics & Automation)

National Taiwan University of Science and Technology

📅 Sep 2014 – Jan 2019 📍 Taipei, Taiwan

- Industrial robotics, mechanical system design, kinematics, CAD/CAM and automated manufacturing.
- Research combining mechanical engineering, computational methods and robotic system implementation.

Master of Design – Mechanical System Design

IIITDM Kancheepuram

📅 Jul 2012 – May 2014 📍 Kancheepuram, India

Mechanical system design, engineering design, automation and product-development methodologies.

Bachelor's Degree in Mechanical Engineering

Visvesvaraya Technological University

📅 Aug 2006 – Jun 2010 📍 Belgaavi, India

Mechanical design, manufacturing and engineering systems.

SELECTED TECHNICAL PUBLICATIONS

Journal Articles

- Bedaka, A.K., Cheng, Y.S., Yen, S.H., Shah, S.H., & Lin, C.Y. (2023). "Trajectory planning method with grinding compensation strategy for robotic propeller blade sharpening application." In: *Journal of Manufacturing Processes*, 86, 294–310.
- Bedaka, A.K., Lee, S.C., Mahmoud, A., & Lin, C.Y. (2021). "A Camera-Based Position Correction System for Autonomous Production Line Inspection." In: *Sensors*, 21(12), 4071.
- Bedaka, A.K., Lin, C.Y., & Huang, S.T. (2019). "Autonomous CAD Model-Based Industrial Robot Motion Planning Platform." In: *International Journal of Robotics and Automation*, 34(4).
- Bedaka, A.K., Vidal, J., & Lin, C.Y. (2019). "Automatic Robot Path Integration Using Three-dimensional Vision and Offline Programming." In: *International Journal of Advanced Manufacturing Technology*, 102(5–8).
- Bedaka, A.K., Mahmoud, A., Lee, S.C., & Lin, C.Y. (2018). "Autonomous Robot-Guided Inspection System Based on Offline Programming and RGB-D Model." In: *Sensors*, 18(11), 4008.

Conference Proceedings

- Bedaka, A. K., & Lin, C. Y. (2020). "CAD-based offline

automation solutions for propeller manufacturing, autonomous inspection and welding offline programming.

- Translated manufacturing requirements into CAD-based software workflows, robot process plans and system-validation methods.
- Bridged mechanical design, manufacturing requirements and robotics software to support industrial automation development.

SELECTED PROJECTS

Industrial Robotic Welding Automation Platform Using 6-DOF Robot and 2-DOF Positioner
CENZ Automation

CAD/CAM OpenCASCADE Industrial Robotics

Developed an integrated CAD-based offline programming solution for automated welding using a 6-DOF industrial robot and 2-DOF positioning system. Implemented geometry processing, robot path generation, kinematic planning and welding-parameter assignment for manufacturing automation.

Autonomous Production-Line Inspection and Position-Correction System
National Taiwan University of Science and Technology

3D Vision Inspection Validation

Developed and experimentally validated an automated inspection system integrating industrial robotics, RGB-D / 3D vision, CAD geometry and position-correction methods for production-line applications.

Automatic Robot Path Generation Using Three-Dimensional Vision and Offline Programming
National Taiwan University of Science and Technology

3D Vision Path Planning Offline Programming

Developed automatic robot path-generation methods integrating three-dimensional vision, CAD information and offline programming for industrial robotic applications.

Robotic Surface Coverage & Motion Planning – UR10e (Ongoing)
Independent Technical Project

ROS2 MoveIt2 NVIDIA Isaac Sim Gazebo

Developing a robotic surface-coverage workflow for painting and sanding applications using a UR10e, integrating coverage-path generation, robot kinematics, motion planning and simulation.

PATENTS & INTELLECTUAL PROPERTY

Welding System and Welding Parameter Optimization Method
Taiwan Patent TW1764130 Issued May 11, 2022
Granted patent covering a welding system and welding-parameter optimization method for automated manufacturing applications.

An Effortless Wheel Replacement Mechanism for Busted Passenger Vehicle/Car Wheels
India Patent Application IN 201641009115 Filed Mar 16, 2016
Patent application for a mechanical mechanism intended to simplify passenger-vehicle wheel replacement.

programming platform for welding applications using 6-DOF and 2-DOF robots.” In: 2020 International Conference on Advanced Robotics and Intelligent Systems (ARIS).

- Bedaka, A. K., & Lin, C. Y. (2018). “CAD-based robot path planning and simulation using OPEN CASCADE.” In: *Procedia Computer Science*, 133, 779–785.

ENGINEERING EXPERTISE

Mechanical System Design Robotics & Automation CAD/CAM
Product Development Prototype Development Mechanical Integration
Engineering Validation Manufacturing Automation Kinematics

ENGINEERING TOOLS

CAD / Mechanical: SolidWorks, Autodesk Inventor, CATIA, OpenCASCADE

CAE / Simulation: ANSYS, Abaqus

Robotics / Automation: ROS, ROS2, MoveIt/MoveIt2, Gazebo, NVIDIA Isaac Sim, ABB RobotStudio, MotoSim

Vision / 3D: OpenCV, Open3D/PCL, RGB-D, Point Clouds

Programming / Data: C++, Python (NumPy, Pandas, SciPy), MATLAB

Engineering Workflow: Git, Docker

APPRECIATION AND AWARDS

Marie Skłodowska-Curie Fellowship
Awarded through the Tecniospring project, Barcelona, Spain. (May 2022)

NTUST Research Scholarship
National Taiwan University of Science and Technology, Taiwan. (Sept. 2014)

Students Mechanism Design Competition
Shortlisted among the top 7 nationally at IIT Roorkee, India. (Dec. 2013)

Pratibha Puraskar
Awarded for securing First Class with Distinction in B.E. (Sep. 2011)

PROFESSIONAL ACTIVITIES

Technical Committee Member · ESCI 2023 – 2025
Reviewer · Sensors, MDPI Reviewer · IJRA, ACTA Press
Invited Lecture · Robotics & Automation, HIT Nidasoshi
Invited Expert Speaker · Bio-ESTaTE, IIITDM Kancheepuram

LANGUAGES

English ● ● ● ● ●
Spanish ● ● ● ● ●
Hindi ● ● ● ● ●